

PROJECT REPORT

ON

“LegalEase: AI-Powered Legal Document Generator”

A Project Report Submitted in Partial Fulfillment of the Requirements for the Degree of

BACHELOR OF COMPUTER APPLICATIONS (BCA)

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CERTIFICATE

This is to certify that the project entitled “LegalEase: AI-Powered Legal Document Generator” is a bonafide work carried out by S Pavithara Register No. 212506777 in partial fulfillment of the requirements for the award of the Bachelor of Computer Applications (BCA) degree.

The project has been completed under my guidance and supervision during the academic year 2026–2027.

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DECLARATION

I hereby declare that the project report entitled “LegalEase: AI-Powered Legal Document Generator” is my original work carried out as part of the Bachelor of Computer Applications programme.

I further declare that this project has not been submitted previously, either fully or partially, to any other institution or university for the award of any degree or qualification.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me in completing the project entitled “LegalEase: AI-Powered Legal Document Generator.”

First, I thank the management and the Department of Computer Applications for providing the opportunity and facilities required to complete this project.

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ABSTRACT

LegalEase is an AI-powered web application designed to simplify the preparation of basic legal document drafts. Traditional document preparation often requires users to understand legal formats, collect information manually and spend considerable time preparing structured documents.

The proposed system provides a user-friendly interface where users can select a document type, enter the required information and generate a structured draft. Artificial Intelligence and Natural Language Processing concepts can be used to organize the information into an appropriate document format.

The system can support document types such as rental agreements, affidavits, simple notices, employment agreements and non-disclosure agreements. The generated document can be previewed and downloaded.

LegalEase is intended as a document-generation and drafting aid. It does not replace professional legal advice, and generated documents should be reviewed by a qualified legal professional before official use.

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CHAPTER 1 – INTRODUCTION

1.1 Background

Legal documents are important in personal, educational, employment and business activities. Preparing such documents manually can be time-consuming because users need to follow a proper structure and provide relevant information.

1.2 Problem Statement

Many users find it difficult to prepare properly structured legal document drafts. Manual preparation may involve repeated typing, formatting errors and difficulty in identifying the information required for a particular document.

1.3 Objectives

- To develop a simple platform for generating structured legal document drafts.
- To reduce the time required for preparing common document formats.
- To collect required information through easy-to-use forms.
- To use AI/NLP concepts for generating readable document content.
- To provide document preview and download functionality.

1.4 Scope

The system focuses on basic document drafting and formatting. It can be extended with additional document templates, multilingual support, document history and professional review workflows.

CHAPTER 2 – EXISTING SYSTEM

2.1 Existing Method

In the existing approach, users may prepare legal documents manually using word processors, search for templates online or seek assistance from professionals. Each method can require time and repeated effort.

2.2 Limitations

- Manual data entry can be time-consuming.
- Users may miss required information.
- Formatting must often be performed manually.
- Searching for suitable templates can be difficult.
- Repeated documents require repeated editing.
- Users may have difficulty understanding document structure.

2.3 Need for Automation

An automated document generator can guide users through the required fields and produce a structured draft. This can improve convenience and reduce repetitive formatting work while keeping the user responsible for verifying the information.

CHAPTER 3 – PROPOSED SYSTEM

3.1 Proposed System

LegalEase provides a centralized interface for selecting a legal document type, entering relevant details and generating a formatted draft. The application can combine predefined templates with AI/NLP processing.

3.2 Main Features

- User registration and login.
- Document type selection.
- Dynamic input forms.
- AI-assisted document generation.
- Document preview.
- Download or save generated documents.
- Basic document history.

3.3 Advantages of the Proposed Approach

The proposed system reduces repetitive formatting work and provides a guided workflow. It can make document drafting more accessible for users who need basic drafts, while clearly communicating that professional legal review may still be necessary.

CHAPTER 4 – SYSTEM REQUIREMENTS

4.1 Hardware Requirements

- Processor: Dual Core processor or above
- RAM: Minimum 4 GB
- Storage: Minimum 10 GB free space
- Display: 1366 × 768 or higher
- Internet connection for AI-based features

4.2 Software Requirements

- Operating System: Windows / Linux / macOS
- Frontend: HTML, CSS, JavaScript or React
- Backend: Python Flask / Django or Node.js
- Database: MySQL / PostgreSQL / SQLite
- AI/NLP service or model
- Web Browser: Chrome, Edge or Firefox

4.3 User Requirements

The user should have basic knowledge of using a web browser and should provide accurate information while filling the document form.

CHAPTER 5 – SYSTEM ANALYSIS

5.1 Functional Requirements

- The system shall allow users to register and log in.
- The system shall display available document types.
- The system shall collect required user inputs.
- The system shall generate a structured document draft.
- The system shall allow users to preview the document.
- The system shall allow users to download or save the document.

5.2 Non-Functional Requirements

- Usability – the interface should be simple and understandable.
- Performance – document generation should be reasonably fast.
- Security – user data should be protected.
- Reliability – the application should handle invalid inputs.
- Maintainability – templates and modules should be easy to update.

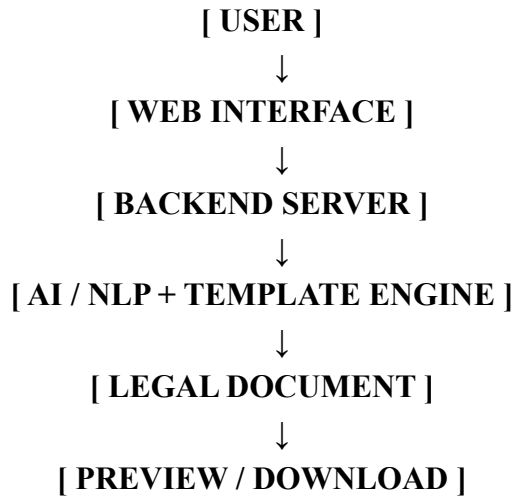
5.3 Feasibility

The project is technically feasible because modern web technologies, databases and AI services can be integrated to create the proposed application. The system can also be developed using commonly available development tools.

CHAPTER 6 – SYSTEM DESIGN

6.1 System Architecture

User → Web Interface → Backend Server → AI/NLP Processing + Document Template Engine → Generated Document → Preview / Download



6.2 Use Case Description

- User logs into the application.
- User selects a document type.
- User enters required details.
- System validates the input.
- AI/template engine prepares the document.
- User previews and downloads the generated draft.

6.3 Data Flow

Input data flows from the user interface to the backend. The backend validates the information and sends relevant content to the template/AI layer. The generated document is returned to the interface for preview and download.

CHAPTER 7 – MODULES

7.1 User Module

Handles registration, login and basic user profile information.

7.2 Document Selection Module

Displays available document categories and allows the user to select the required document.

7.3 Input Module

Collects details such as names, addresses, dates, terms and other document-specific information.

7.4 AI Document Generation Module

Processes the supplied information and creates a structured draft using predefined templates and/or AI/NLP techniques.

7.5 Preview Module

Displays the generated document before saving or downloading.

7.6 Download Module

Allows the user to download the generated document in a suitable format.

7.7 History Module

Stores basic document generation history so users can access previously created drafts.

CHAPTER 8 – TECHNOLOGIES USED

8.1 Frontend

HTML provides the page structure, CSS provides styling and JavaScript handles interactive features. React may be used for a component-based interface.

8.2 Backend

Python Flask/Django or Node.js can be used to manage requests, validation, authentication and document generation.

8.3 Database

SQLite, MySQL or PostgreSQL can store user accounts, document templates, input information and document history.

8.4 Artificial Intelligence / NLP

AI/NLP can be used to transform structured user information into readable document text, identify relevant fields and assist with template selection.

8.5 Document Generation

A document-generation library can be used to create downloadable DOCX or PDF files from the generated content.

CHAPTER 9 – DATABASE DESIGN

9.1 Main Tables

Table	Field	Type	Purpose
Users	user_id	Integer	Unique user ID
Users	name	Varchar	User name
Users	email	Varchar	Login email
Documents	document_id	Integer	Document ID
Documents	document_type	Varchar	Type of document
Documents	created_at	DateTime	Creation date
Documents	content	Text	Generated draft

9.2 Database Relationship

One user can generate multiple documents. Therefore, the Users table can have a one-to-many relationship with the Documents table through user_id.

Users (1) ————— (M) Documents

CHAPTER 10 – IMPLEMENTATION

10.1 Implementation Steps

- Create the project environment and database.
- Develop registration and login screens.
- Create document selection interface.
- Create dynamic forms for each document type.
- Implement backend validation.
- Integrate template and AI/NLP processing.
- Create document preview functionality.
- Implement download and history features.
- Test the complete application.

10.2 Sample Workflow

For example, when the user selects a Rental Agreement, the system asks for landlord details, tenant details, property information, rental period and other required fields. After validation, the system places the information into the selected document structure and generates a draft.

10.3 Security Considerations

Authentication, password protection, input validation and controlled access should be implemented. Sensitive user information should be handled securely and should not be unnecessarily exposed.

CHAPTER 11 – TESTING

11.1 Testing Objective

Testing ensures that each module performs the expected function and that invalid or incomplete input is handled appropriately.

Test Case	Input / Action	Expected Result	Status
TC01	Valid login details	User successfully logs in	Pass
TC02	Invalid login details	Error message displayed	Pass
TC03	Select document type	Correct form displayed	Pass
TC04	Submit required fields	Document draft generated	Pass
TC05	Missing required field	Validation message displayed	Pass
TC06	Preview document	Generated content displayed	Pass
TC07	Download document	File downloaded successfully	Pass

11.2 Testing Types

The project can use unit testing, integration testing, system testing and user acceptance testing.

CHAPTER 12 – RESULTS AND SCREENSHOTS

The proposed LegalEase application provides a guided workflow for generating structured legal document drafts. After entering valid information, the user can view the generated content and download it.

12.1 Expected Screens

- Login / Registration Screen
- Dashboard Screen
- Document Selection Screen
- Document Input Form
- AI Generation / Processing Screen
- Document Preview Screen
- Download / History Screen

[SCREENSHOT PLACEHOLDER – LOGIN PAGE]

[SCREENSHOT PLACEHOLDER – DOCUMENT SELECTION]

[SCREENSHOT PLACEHOLDER – GENERATED DOCUMENT PREVIEW]

Actual screenshots can be inserted here after implementing the application.

CHAPTER 13 – ADVANTAGES, LIMITATIONS & FUTURE SCOPE

13.1 Advantages

- Simple and user-friendly workflow.
- Reduces repetitive document formatting.
- Provides structured document drafts.
- Can support multiple document templates.
- Allows quick preview and download.

13.2 Limitations

- Generated documents may require professional legal review.
- Legal requirements can vary by jurisdiction.
- AI-generated text may require verification.
- Internet connectivity may be required for some AI services.

13.3 Future Enhancements

- Support for more legal document types.
- Multilingual document generation.
- Electronic signature integration.
- Document version control.
- Professional lawyer review workflow.
- Improved privacy and secure document storage.

CHAPTER 14 – CONCLUSION AND REFERENCES

14.1 Conclusion

LegalEase demonstrates how web technologies and Artificial Intelligence can be combined to simplify the preparation of basic legal document drafts. The system provides a guided process for selecting a document, entering information, generating structured content and downloading the result.

The project can be further enhanced with additional templates, multilingual support, stronger security and professional review features. LegalEase is designed as a drafting aid and should not be treated as a replacement for qualified legal advice.

14.2 References

- Python Documentation – <https://docs.python.org/>
- MDN Web Docs – <https://developer.mozilla.org/>
- React Documentation – <https://react.dev/>
- MySQL Documentation – <https://dev.mysql.com/doc/>
- Natural Language Processing – standard academic references and documentation
- Artificial Intelligence concepts – standard textbooks and academic resources

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